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Evaluation of MR-compatibility of Electrostatic Linear Motor

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***Abstract* – This paper reports a new MRI (Magnetic Resonance Imaging) compatible actuator that can operate near an MRI without degrading its imaging quality. A high-power electrostatic motor is introduced and its MR-compatibility is examined, both experimentally and theoretically. The proposed motor was fabricated with only paramagnetic materials, and its MR-compatibility was evaluated by measuring the output performance of the motor and the signal-to-noise ratios of the MRI images. The experimental results showed that motor's thrust force while positioned within the MRI is almost the same as that in normal operation. The signal-to-noise investigation indicated that the motor did not affect the MRI images when it was operated over 60 cm away from the MRI's RF coil. The theoretical calculations showed that the magnetic field generated by the motor is relatively small and the effect of the MRI's magnetic fields on the motor is negligible, suggesting that the motor is MR-compatible.**

Index Terms – fMRI, MR-compatible, Electrostatic Motor, Actuator, Linear Motor

I. Introduction

MRI (Magnetic Resonance Imaging) has established itself as a standard in medical diagnostics and neuroscience because it provides many types of functional tomographic images while avoiding exposure to radiation. Recently, robotic technologies that utilize MRI have attracted increased interest. In medicine, surgical robots guided by real time 3D MRI can perform precise, minimally invasive operations that require only minimal recovery time [1]. In neuroscience, a functional MRI (fMRI) with a robotic interface allows for the advanced investigation of brain mechanisms that are used to perform tasks that require arbitrary dynamics [2].

There are, however, significant challenges for robots working within an MRI environment [3]. The extraordinary magnetic field produced by the MRI prevents the use of conventional electromagnetic actuators, which could easily be attracted by the field and could possibly lead to fatal accidents. Moreover, magnets and the ferromagnetic materials used in the actuators interfere with the MRI's magnetic field and scanning radio frequency (RF) signals, causing artifacts in the MR images.

To overcome these difficulties, non-magnetic actuators, such as ultrasonic motors [4] and electrostrictive polymer actuators (EPAM) [5], were introduced as viable alternatives to the conventional electromagnetic actuators. Among these, ultrasonic actuators have been primarily incorporated by previous studies. For example, they have been employed in the manipulators of MRI-guided surgical robots [6], which have exhibited good MR-compatibility in some respects. Ultrasonic actuators, however, are not fully MR-compatible because they disturb MRI imaging if they are operated while the MRI is scanning images [1].

This study focuses on another non-magnetic actuator, the electrostatic actuator. Electrostatic actuators are well known in the field of MEMS [7-9], and most of the research in the past has focused on incorporating electrostatic actuators into micro actuators. This is due to the size effect of the electrostatic force that allows the electrostatic force to be dominant in a micro region. On the other hand, it has also been reported that large and strong electrostatic motors can be realized by combining a large number of micro actuator elements [10-11]. An example of such an electrostatic actuator is the *Dual Excitation Multiphase Electrostatic Drive* (DEMED) [11]. In reported studies, some DEMED prototypes exhibited superior output performance, including maximum speeds of over 1m/s and thrust forces of over 300 N [11]. And because this motor does not require an electromagnetic field for operation and does not contain ferromagnetic materials, it is far less likely to interact with the surrounding electromagnetic fields than electromagnetic motors. In addition, the electrostatic motor operates at lower frequencies (typically, up to 1kHz) and at lower currents than ultrasonic actuators, so it should interfere even less with MRI imaging when it operates.

This paper evaluates a prototype electrostatic motor that is fabricated with paramagnetic materials. Its MR-compatibility in fMRI environments is tested in terms of thrust force generation capability and signal-to-noise ratio of the MRI images. Numerical calculations are also provided to theoretically determine the interactions between the electrostatic motor and a surrounding magnetic field, in order to demonstrate its potential MR-compatibility.

II. High-Power Electrostatic Film Motor

The basic structure of the DEMED is shown in Fig. 1. In its simplest configuration, the motor consists of a pair of thin plastic films that are manufactured using flexible printed circuit (FPC) technology. Each film contains three-phase parallel electrodes that are aligned at regular 200 μm

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intervals. Glass beads with diameter of 20 μm are inserted between the films to reduce friction.

Figure 2 illustrates the driving principle for the motor. The motor is a synchronous linear motor driven with a high three-phase AC voltage. As depicted in the figure, a three-phase AC voltage source is connected to the three-phase electrodes on the both the stator and the slider, but with the connection order reversed. When the three-phase AC is applied, the traveling voltage distributions on both the stator and the slider are excited, causing them to move in opposite directions at speeds of $\pm u_d$. These voltage distributions induce an electrostatic interaction that maintains a stable spatial phase difference between the two. As a result, the slider runs at a speed of $2u_d$ in the synchronous state so that relative speed between two voltage distributions becomes zero. The slider speed u can be expressed as:

$$u = 2u_d = 6pf \quad (1)$$

where p is the electrode pitch (200 μm) and f is the frequency of the applied AC voltage.

The thrust force of the motor is proportional to the amplitude of the applied AC squared. The lowest functional applied voltage is around 500 $V_{0\text{-}p}$. However, a voltage higher than 1 $kV_{0\text{-}p}$ is necessary to obtain a practically large force. When the applied voltage becomes sufficiently high, electric discharges occur in the air around the motor, which disturbs its electrostatic field and causes improper motor behavior. To prevent these discharges, the motor is usually immersed or dipped in a dielectric liquid like Fluorinert (3M) or silicon oil.

Practical configurations stack several pairs of films so that the motor can generate a larger force; the output force capability is directly proportional to the number of stacked pairs. A stacked motor configuration is illustrated in Fig. 3.

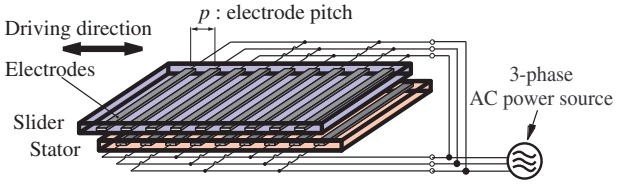


Fig. 1 Basic structure of the DEMED

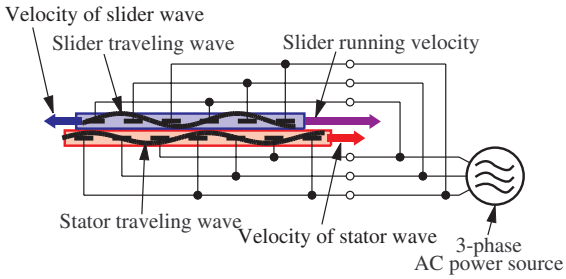


Fig. 2 DEMED driving principle

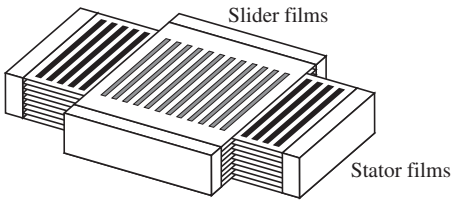


Fig. 3 Stacked motor

III. Theoretical Evaluation

This section provides a theoretical examination of the proposed motor's MR-compatibility with respect to two viewpoints, which correspond to the two experiments conducted in later sections. The first is the effect of the MRI magnetic field on the motor's output force capability and the second is the effect of the motor's magnetic field on the MRI. For the purposes of this discussion, the motor is assumed to consist of only one pair of slider and stator films (no stacking), the stator has the same structure and the same size as the slider, and the gap between the slider and the stator is negligible (meaning the slider and the stator occupy the same location). The motor is also assumed to be operating at a constant speed.

Figure 4 shows the coordinate system used in this chapter. The coordinate system, called the *motor coordinate system*, has its origin at the center of the motor. The *x-axis* is defined along the motor's operation direction, the *y-axis* along the lateral direction, and the *z-axis* along the normal direction.

A, *Effect of the magnetic field on motor operation*

A surrounding magnetic field can affect the slider's motion in two different ways. It can attract paramagnetic materials within the motor and it can generate Lorenz forces on the currents flowing in the slider. Because the motor contains only minute quantities of materials with magnetic susceptibilities, the former effect is negligible. Therefore, the effect of the Lorenz force is primarily discussed.

When currents flow inside a slider that is placed in a magnetic field, the currents induce a Lorenz force on the slider. The total Lorenz force on the slider can be calculated if the current distribution in the slider is known. The electrodes of the slider film, which are located on both sides of the film and are schematically illustrated in Fig. 4, are a combination of three-phase parallel electrodes and feeding electrodes (α, β, γ_1, γ_2). Due to spatial limitations, the third-phase electrodes are connected alternately to two feeding electrodes (γ_1, γ_2) while the electrodes for the other two phases are connected to identical electrodes (α, β),

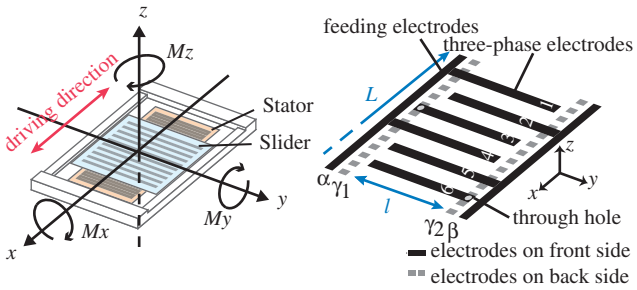


Fig. 4 Schematic illustration of the electrodes and the coordinate system for the theoretical evaluation

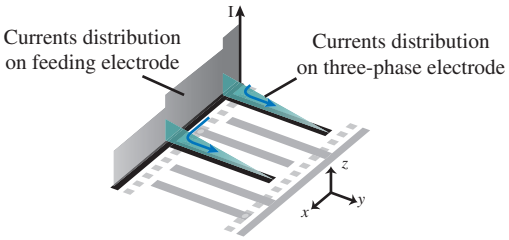


Fig. 5 Current distribution in the electrodes

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respectively. Therefore, this *three*-phase electrode pattern repeats every *six* electrodes.

Since the operating frequency of this motor is relatively low (typically, up to 1 kHz) and the transitional signal reflections and propagations can be neglected, the current distribution in each three-phase electrode can be estimated as depicted in Fig. 5, or

$$\mathbf{i}_n(y)=\begin{cases}\left[0,\ i_0\left(\frac{l}{2}-y\right)\sin(\omega t),0\right] & (n=3k+1;k=0,1,2\ldots)\\ \left[0,i_0\left(y-\frac{l}{2}\right)\sin\left(\omega t+\frac{2\pi}{3}\right),0\right] & (n=3k+2)\\ \left[0,\ i_0\left(y-\frac{l}{2}\right)\sin\left(\omega t-\frac{2\pi}{3}\right),0\right] & (n=6k)\\ \left[0,\ i_0\left(\frac{l}{2}-y\right)\sin\left(\omega t-\frac{2\pi}{3}\right),0\right] & (n=6k+3)\end{cases} \quad (2)$$

where n is the electrode number and i_0 is the current amplitude at the root of each three-phase electrode, which is dependent on voltage and frequency. The current in a feeding electrode comes from one edge and branches into three-phase electrodes, as shown in Fig. 5. The current distribution in the feeding electrodes can be written using a linear function approximation as:

$$\begin{cases}\mathbf{i}_\alpha(x)=\left[I_0\left(\frac{L}{2}-x\right)\sin(\omega t),\ \ 0,\ \ 0\right]\\ \mathbf{i}_\beta(x)=\left[I_0\left(\frac{L}{2}-x\right)\sin\left(\omega t+\frac{2\pi}{3}\right),\ \ 0,\ \ 0\right]\\ \mathbf{i}_{\gamma 1}(x)=\mathbf{i}_{\gamma 2}(x)=\left[\frac{I_0}{2}\left(\frac{L}{2}-x\right)\sin\left(\omega t-\frac{2\pi}{3}\right),\ \ 0,\ \ 0\right]\end{cases} \quad (3)$$

where I_0=N i_0 / 3 and N is the total number of the three-phase electrodes. Therefore, the total current distribution **I** in the slider is

$$\mathbf{I}(x,y)=\left[I_x,I_y,I_z\right]=\mathbf{i}_\alpha+\mathbf{i}_\beta+\mathbf{i}_{\gamma 1}+\mathbf{i}_{\gamma 2}+\sum_1^N\mathbf{i}_n \quad (4)$$

Using the current distribution **I**, the total Lorenz force **F** under a static magnetic field with magnetic flux density **B** can be calculated using an integral over entire film area S:

$$\mathbf{F}=\int_S\mathbf{I}(x,y)\times\mathbf{B}\,dxdy \quad (5)$$

Since the total amount of three-phase current is always zero, Equation (5) always gives zero, meaning that the magnetic field does not generate any translational force on the slider.

On the other hand, rotational moments due to the Lorenz force do not become zero. The moment $\mathbf{M}$ = [M_x, M_y, M_z] can be calculated as:

$$\mathbf{M}=\int_S\mathbf{I}(x,y)\times\mathbf{B}\times\mathbf{r}(x,y)\,dxdy \quad (6)$$

where $\mathbf{r}(x, y)$ is a position vector. With this equation, the pitching moment (M_y) is always zero, whereas rolling (M_x) and yawing (M_z) moments can be generated by a surrounding magnetic field. It is also discovered that B_z does not affect M_x and that B_x and B_y do not affect M_z. The numerically calculated M_x and M_z are plotted in Fig. 6 for normalized B_x, B_y and B_z. In this calculation, the number of three-phase electrodes was set to 499, which is the number used in the prototype motor described in a later section. The current amplitude of each phase, I_0, was normalized to 1 A.

Since the base plate of the housing constricts the pitching and rolling motions of the prototype slider, only a yawing moment could affect the motor's operation. However, when practical conditions **B** = [0, 0, 1.5T] and I_0 = 5 mA are used, the yawing moment is calculated to be 0.33 μNm. This moment is a very small value, so it can be concluded that the effect of the MRI magnetic field on motor operation is negligible.

B, Magnetic fields generated by the motor

One possible cause of noises in MRI images could be the magnetic field generated by the motor's currents. The total magnetic field from the motor can be calculated from the total current in the motor, including the slider and stator. The total motor currents $\mathbf{I}_{total}(x, y)$ can be written as $\mathbf{I}_{total}(x, y) = 2\mathbf{I}_0(x, y)$, since both the structure and position of the stator are assumed the same as the slider. Then, the total magnetic flux density at a point $(x, y, z) = (x_0, y_0, z_0)$ can be calculated using *Biot-Savart's* law as;

$$\mathbf{B}(x_0,y_0,z_0)=\int_S\mu_0\frac{\mathbf{I}_{total}(x,y)\times\mathbf{r}(x_0,y_0,z_0)}{4\pi\left|\mathbf{r}(x_0,y_0,z_0)\right|^3}dxdy \quad (7)$$

The flux density **B** was numerically calculated and is plotted in Figs. 7 and 8. Figure 7 shows the magnetic field at (x, y, z) = (0, 1m, 0) when current amplitude of each phase is normalized to 1A. The field drastically varies with the phase of the applied AC voltage, with the ratio between the maximum and the minimum values reaching 10^3. The variation in the maximum values along all three axes is plotted in Fig. 8. The plot reveals that the magnetic field

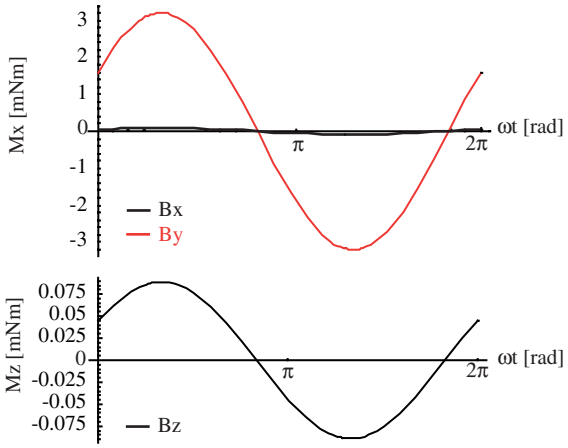


Fig. 6 Relationships between the time and moments when the direction of the magnetic field is along each axis. The intensity of the magnetic field is 1 T.

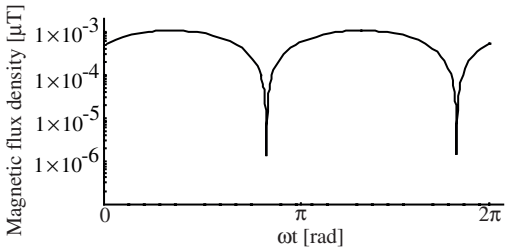


Fig. 7 Total amount of magnetic flux density at (x, y, z) = (0, 1m, 0) with normalized current of 1A.

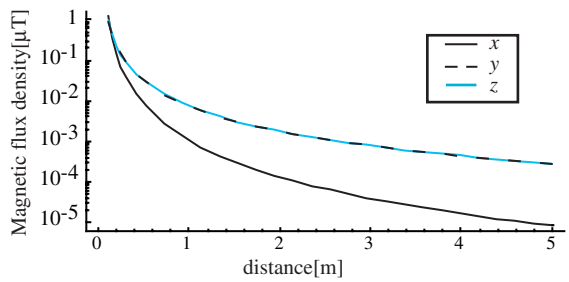


Fig. 8 Variation of the maximum value of the magnetic flux density along each axis with normalized current of 1 A.

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decreases rapidly as the distance from the motor increases. This is because the magnetic fields from the three phases are cancelled at farther points.

Using the conditions employed by the following experiments (phase current: 5 mA, position: x = 35 cm; y = 0; z = 0), the maximum magnetic flux density, which is almost the same as the field fluctuation, is calculated to be 0.55 nT. This value is far smaller than the MRI field homogeneity, suggesting that it has no significant effect on MRI imaging.

IV. Experimental Setup for MR-compatiblity tests

A, fMRI

An fMRI (Marconi, MAGNEX ECLIPSE 1.5 T), whose specifications are summarized in Table 1, was used for the MR-compatibility tests. The magnetic flux density of the static magnetic field was 1.5 T, and the homogeneity of the field was less than 1 ppm/50 cm DSV. A field echo (FE) sequence was used as the imaging sequence for the experiments.

B, Nonmagnetic prototype motor

The nonmagnetic prototype shown in Fig. 9 was used for the MR-compatibility tests. The prototype consists of only paramagnetic materials, with the housing and screws fabricated from acrylic resin. Copper was used for the electrode and polyimide was used for the base and cover layers of the stator and slider films. The inside of the housing was filled with dielectric liquid (Fluorinert FC-77, 3M).

The stator and slider films for the prototype are shown in Fig. 10. The prototype contains two pairs of stacked stator and slider films. Although the prototype does not include a linear guide, the direction of the slider motion was constricted by the electrostatic force.

Table 1 Specification of fMRI

Static magnetic field [T]	1.5
Homogeneity of magnetic field	<1ppm/50cmDSV
Gradient magnetic field [mT/m]	27
Switching rate [mT/m/ms]	72
RF coil Power [kW]	20

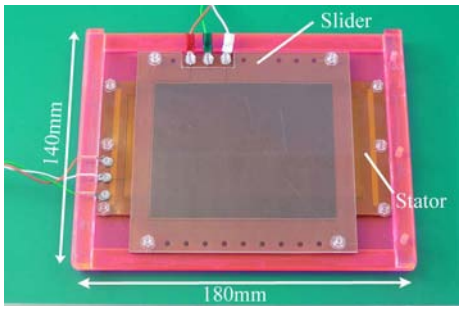


Fig. 9 Nonmagnetic prototype motor (height: 10mm)

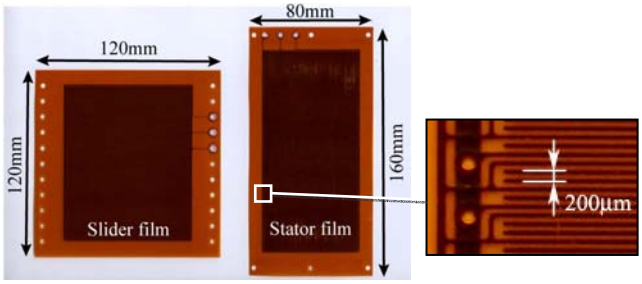


Fig. 10 Electrode films

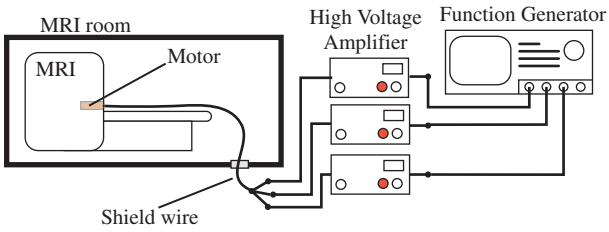


Fig. 11 Measurement configuration

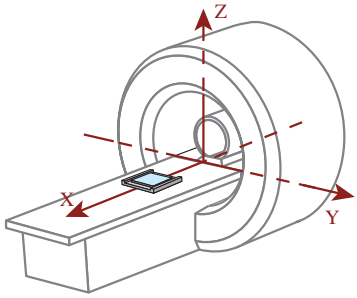


Fig. 12 MRI-coordinate system

C, Measurement configuration

Figure 11 shows the configuration of the motor and its power sources. The motor was fixed to the fMRI bed, while the power sources, three high voltage amplifiers (AS-123, NF corp.), were located outside the MRI room. The motor and the power sources were connected using an 8 m shielded three-wire cable. The function generator (AG-1200, YOKOGAWA), also located outside the MRI room, generated a three-phase driving AC signal that was fed to the high-voltage amplifiers to be thousand-folded.

It was verified that there was no discernible magnetic attraction force on the prototype when the motor was operating inside the MRI gantry, which suggests that the proposed motor is *MR safe* [3].

V. Experimental Evaluation of MR-compatibiliity

In this study, MR-compatibility was evaluated with respect to two points: the influence of MRI's strong magnetic field and the influence of motor's operation on MRI imaging. The former was evaluated through thrust force measurements, while the latter was evaluated through the signal-to-noise ratio (SNR) of the MRI images.

The *MRI coordinate system* defined in Fig. 12 is used to describe the measurement conditions. The origin of the system is set at the edge of the RF-coil that was positioned at the center of the gantry. The X-*axis* is defined as the direction parallel the direction of the MRI's static magnetic field.

A, Thrust force comparison

Since the materials used in the prototype are paramagnetic, the prototype is not affected by the surrounding magnetic field when the motor is not operating. However, when the motor is operating, the surrounding magnetic field induces a Lorenz force on electric current in the slider, as was discussed in the theoretical evaluation section. Although the theoretical calculations indicated that the influence is negligible, this result is verified in the following experiment.

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To evaluate the influence, the motor's thrust force was measured under three different conditions: without magnetic fields, inside the MRI gantry with the motor operating along X-axis, and in the MRI gantry with the motor operating along the Y-axis. Since the MRI's strong magnetic field prevents the use of conventional force sensors, the thrust force was measured using a paramagnetic blade spring, which is made from phosphor bronze. The slider pulled the blade using a string and the thrust force was estimated from the maximum displacement of the spring. Figure 13 shows the results of the thrust force measurements. No significant difference is evident from these three data series, which suggests that the effect of surrounding magnetic field is negligible. The maximum thrust force was found to be between 9 to 10 N at a voltage amplitude of 1.6 $kV_{0\text{-}p}$.

B, Evaluation based on the image SNRs

The investigation calculated the SNRs using the axial images (images on the YZ-plane) of a cylindrical phantom, which was a plastic cylindrical vessel filled with a $CuSO_4$ and $NiCl_2$ solution in water. Typical images of the phantom under various SNRs are shown in Fig. 14. Each image was obtained as an average of five consecutive images over a period of 30 seconds. The SNR for each image was calculated as the ratio of the mean signal value at the center of the image (signal) against the mean signal value of the four corners (noise). The intrinsic SNR of the phantom, which was measured without the motor, was 42 dB. When the motor with the shielded cable was positioned 35 cm from the RF coil (X = 35 cm, Y = Z = 0 cm) and not energized, the SNR was calculated to be 41 dB.

(a) SNR vs. applied voltage

First, the variation in SNR with respect to the applied voltage was measured. During the measurements, the motor was positioned at (X, Y, Z) = (35 cm, 0 cm, 0 cm) within the gantry and driven in either the X or Y direction. The motor was driven with an applied frequency of 20 Hz, which corresponded to a synchronized speed of 24 mm/s.

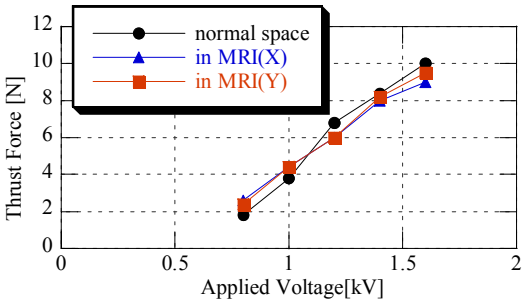


Fig. 13 Thrust force comparison
The motor was positioned in an ordinary space and in the MRI.

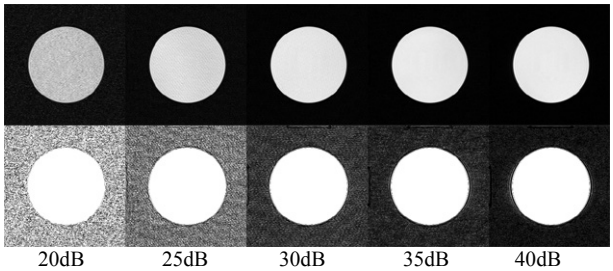


Fig. 14 Typical images of the phantom under various SNRs.
Upper images are raw images. Lower images are contrast-emphatic images. These images were taken in the following experiments.

Each image was obtained over 30 seconds, during which the motor reciprocated with a 2.4 cm stroke. The reciprocation direction was reversed by switching two of the three AC phases. The resulting voltage and current waveforms, both measured for one phase, are plotted in Fig. 15.

Figure 16 shows relationship between the applied voltage amplitude and the SNR of the imaging results. The labels X and Y denote the direction along which the motor moved. When the voltage amplitude is less than 1 kV, the SNR was not degraded. However, amplitudes over 1.2 kV resulted in a proportionally lower SNR.

(b) SNR vs. driving frequency

Next, SNR variation with respect to driving frequency (motor speed) was measured. For these experiments, the motor was positioned at (X, Y, Z) = (35 cm, 0 cm, 0 cm) and operated in the X direction with a motion stroke of 2.4 cm. The amplitude of the applied voltage was 1.2 $kV_{0\text{-}p}$. Figure 17 summarizes the imaging results. No significant difference was found between the SNRs of the images.

Since the motor is a capacitive load, a higher frequency results in higher current, which could generate a larger magnetic field. The measured relationship between the phase current amplitude and the applied frequency is plotted in Fig. 18. The figure shows that the current amplitude is nearly proportional to the frequency. Nevertheless, Fig. 17 indicates that the SNR is almost constant within this

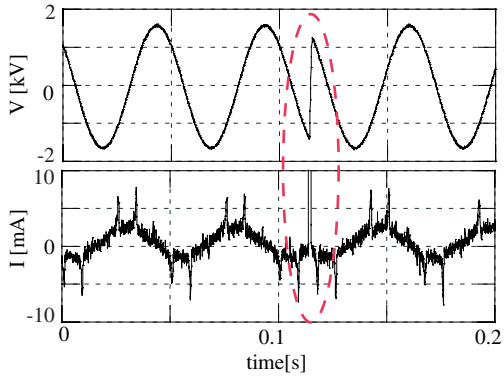


Fig. 15 Voltage and current for one of three phases of the motor.
The motor changed its operating direction at around 0.12 seconds.

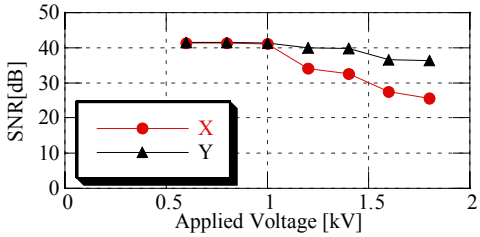


Fig. 16 Relationship between SNR and applied voltages.
The motor was positioned in the gantry, 35 cm from the RF-coil.

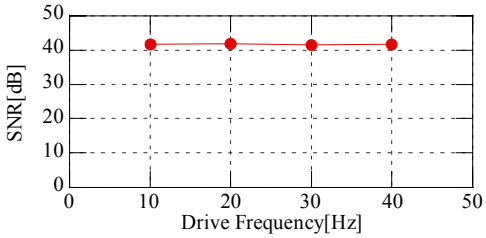


Fig. 17 Relationship between driving frequency and SNR.
Applied voltage was 1.2 $kV_{0\text{-}p}$. The motor was positioned in the gantry, 35 cm from the RF-coil.

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frequency region. This result suggests that the effects of magnetic fields generated by these currents are negligibly small, verifying the theoretical results.

(c) SNR vs. motor location

Finally, the relationship between SNR and motor location was investigated. For this experiment, the motor was positioned at four points: X = 30, 60, 90, and 120 cm, with Y = Z = 0 cm. The motor operated along either the X or Y direction with a motion stroke of 2.4 cm. The applied voltage amplitude was 1.4 $kV_{0\text{-}p}$ and the driving frequency was 20 Hz (24 mm/s).

The results are shown in Fig. 19. Regardless of the driving direction, the motor caused slight noise at X = 30 cm. On the other hand, no significant SNR degradation was found when the motor was positioned more than 60 cm from the RF coil.

C, Discussion on SNR measurements

The measurements from these experiments discovered that the higher the applied voltage or the closer the motor is to the RF-coil, the lower the resultant SNR. However, the SNR proved to be independent of the driving frequency and, therefore, the operation speed, within a certain operation range. Since the amplitude of the current is proportional to the driving frequency, the results also suggest that the amplitude of electric current is not the dominant parameter for SNR in this parameter region.

The theoretical calculations predicted that the operation current is too small to affect MRI imaging. However, the motor was reciprocating during MRI scanning in the experiment, which causes voltage jumps, as shown in Fig. 15. These voltage jumps cause current spikes that have relatively larger amplitudes and frequencies than during normal operation of the motor. Since the degree of voltage jump is proportional to the operating voltage, the large operating voltage causes a large current spike. Therefore, because the SNR of the MRI image is dependent upon the voltage amplitude and not the driving frequency, it can be concluded that the major cause of MRI noise is a current spike caused by a voltage jump. These voltage jumps can be minimized by improving the motion profile, which will be tested in future works.

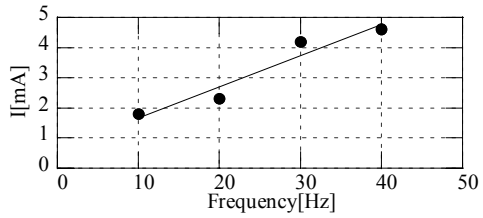


Fig. 18 Relationship between the amplitude of the phase current and the driving frequency

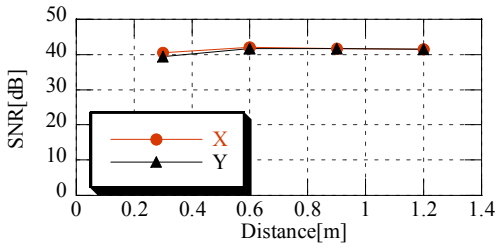


Fig. 19 Relationship between distance and SNR.
Applied voltage was 1.4 $kV_{0\text{-}p}$, and the driving frequency was 20Hz.

VI. Conclusion

This paper evaluated the MR-compatibility of a high-power electrostatic linear motor called the *Dual Excitation Multiphase Electrostatic Drive*. Theoretical analyses based on the motor's current distribution were conducted. The calculations showed that the strong MRI magnetic field has almost no effect on the operation of the motor. The analyses also showed that magnetic flux density generated by the motor is sufficiently small compared to the homogeneity of the MRI's static magnetic field that the motor operation does not affect MRI imaging.

A prototype motor comprised of paramagnetic materials was fabricated to experimentally evaluate MR-compatibility. The results showed that the prototype motor could operate properly in the MRI environment without significant effects to its thrust force generation. The motor was also found to not affect MRI imaging when it was positioned more than 35 cm away from the RF-coil and operated with applied voltages less than 1 kV. When the applied voltage is higher than 1 kV, however, noise was occasionally observed on MRI images. This noise is likely due to spike currents generated when the motor was reversed. They can be eliminated by improving the motion profile of the motor, which will be addressed in future works.

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